LLM PREPROCESSING COMPARISON REPORT

Baseline vs Preprocessed Model Performance Analysis

Experiment ID	319852e17369
Generated at	2026-11-09 14:36:04
Total records	10
Models tested	5
Queries tested	1
Successful responses	10
Failed responses	0

1. Model-wise Performance Summary

Model	Condition	Total	Successful	Failed	Avg latency (s)	Avg response chars
gemma	baseline	1	1	0	13.975	557.0
gemma	preprocessed	1	1	0	10.474	564.0
llama	baseline	1	1	0	14.023	625.0
llama	preprocessed	1	1	0	18.146	622.0
qwen	baseline	1	1	0	9.652	676.0
qwen	preprocessed	1	1	0	6.626	707.0
smollm2	baseline	1	1	0	10.417	597.0
smollm2	preprocessed	1	1	0	14.491	553.0
tinyllama	baseline	1	1	0	8.193	644.0

Model	Condition	Total	Successful	Failed	Avg latency (s)	Avg response chars
tinyllama	preprocessed	1	1	0	10.852	625.0

2. Baseline vs Preprocessed Comparison

Model	Paired queries	Baseline latency	Preprocessed latency	Latency change	Baseline chars	Preprocessed chars	Length chang
gemma	1	13.975	10.474	-3.501s	557.0	564.0	+7.0
llama	1	14.023	18.146	+4.123s	625.0	622.0	-3.0
qwen	1	9.652	6.626	-3.026s	676.0	707.0	+31.0
smollm2	1	10.417	14.491	+4.074s	597.0	553.0	-44.0
tinyllama	1	8.193	10.852	+2.659s	644.0	625.0	-19.0

Successfully paired model-query comparisons: 5. Negative latency change means the preprocessed condition was faster. Positive latency change means it was slower. A negative response-length change means the response was shorter.

3. Failed Requests and Error Details

No failed requests were recorded.

4. Interpretation and Evaluation Notes

- Latency values are averaged over successful responses only.
- Failed requests are reported separately and are not treated as zero-character successful responses.
- Baseline and preprocessed comparisons are paired using the same model and original query.
- Response length measures the number of characters returned; it does not establish answer correctness.
- A shorter response or lower latency does not automatically mean better response quality.
- Quality evaluation requires a documented rubric or validated benchmark. No factual accuracy score is inferred by this report.